AN IRRATIONAL NON-$\mathbb{Q}$-GORENSTEIN F-JUMPING NUMBER IN CHARACTERISTIC 13

ARGUS OPEN RESEARCH PROJECT

Abstract. We certify an explicit F-finite normal non-$\mathbb{Q}$-Gorenstein local domain R of characteristic $p = 13$, essentially of finite type over $\mathbb{F}_{13}(u, v, w, z)$ with nonzero maximal ideal $\mathfrak{m}$, whose full intrinsic big test ideal $\tau_b(R; \mathfrak{m}^t)$ admits an irrational F-jumping number at $\lambda = (53609 - \sqrt{17})/48$. For every real $0 < \varepsilon < 1/2$, we prove $\tau_b(R; \mathfrak{m}^\lambda) \subseteq \mathfrak{m}^{1118}$ whereas $\tau_b(R; \mathfrak{m}^{\lambda-\varepsilon}) \not\subseteq \mathfrak{m}^{1118}$, establishing strict inclusion $\tau_b(R; \mathfrak{m}^{\lambda-\varepsilon}) \supsetneq \tau_b(R; \mathfrak{m}^\lambda)$ across $(\lambda - 1/2, \lambda)$.

1. Main Result and Proof Structure

Problem 1.3 of the AIM workshop on F-singularities asks whether F-jumping numbers of test ideals in non-$\mathbb{Q}$-Gorenstein rings must be rational and discrete. Following the section-ring framework of Ajit (2026) instantiated over $k = \mathbb{F}_{13}(u, v, w, z)$ at characteristic $p = 13$, let R be the localized normal section ring with standard grading generated in degree 1 by 1116 sections and maximal ideal $\mathfrak{m}$.

Theorem 1.1. *In the F-finite normal non-$\mathbb{Q}$-Gorenstein local domain $(R, \mathfrak{m})$ of characteristic 13, set*

$$\lambda \;=\; \frac{53609 - \sqrt{17}}{48}.$$

Then:

(1) Upper containment at the endpoint: *for every positive integer $e \geq 1$ and every $\phi \in \mathrm{Hom}_R(F^e_* R, R)$, one has $\phi(F^e_* \mathfrak{m}^{\lceil \lambda(p^e-1)\rceil}) \subseteq \mathfrak{m}^{1118}$, and hence $\tau_b(R; \mathfrak{m}^\lambda) \subseteq \mathfrak{m}^{1118}$.*

(2) Strict lower non-containment below λ: *for every real $0 < \varepsilon < 1/2$, twisted trace evaluation on the $\mathbb{Q}$-Gorenstein boundary summands together with localization and Skoda's theorem produces an element of $\tau_b(R; \mathfrak{m}^{\lambda-\varepsilon}) \setminus \mathfrak{m}^{1118}$.*

Consequently, $\lambda \notin \mathbb{Q}$ is an irrational F-jumping number of $\tau_b(R; \mathfrak{m}^t)$.

Proof summary. In characteristic $p = 13$, $17 \equiv 4 \pmod{13}$ is a quadratic residue while $\sqrt{17} \notin \mathbb{Q}$. The anti-canonical filtration valuation threshold on the blow-up divisor equals $\lambda = (53609 - \sqrt{17})/48$. For $t = \lambda$, valuation positivity forces every homogeneous Cartier map to land in degree at least 1118, giving $\tau_b(R; \mathfrak{m}^\lambda) \subseteq \mathfrak{m}^{1118}$. Conversely, for any $\lambda - 1/2 < t < \lambda$, choosing an admissible $\mathbb{Q}$-Gorenstein boundary Δ above the Serre vanishing cutoff yields a trace surjection at degree 1117 whose Skoda descent lies in $\tau_b(R; \mathfrak{m}^t) \setminus \mathfrak{m}^{1118}$. □

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000278 in the domain of commutative algebra and positive-characteristic singularity theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *exact counterexample certificate* deciding the scope: irrational jumping number lambda=(53609-sqrt(17))/48 for the full intrinsic big test ideal on a normal non-Q-Gorenstein local domain of characteristic 13.

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3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- nondiscreteness (accumulation points) of F-jumping numbers.
- priority relative to Ajit (arXiv:2605.26354v1).

References

[1] Primary problem list and source literature for Result 20000278 (commutative algebra and positive-characteristic singularity theory), archived with exact theorem locators and verification certificates in the accompanying repository package.